

# Spring AI RAG Sample Document

Spring Boot is a Java framework for building production-ready applications.

Dependency Injection is a design principle that reduces coupling between components.

Microservices split applications into small independently deployable services.

Retrieval-Augmented Generation (RAG) combines document retrieval with a Large Language Model.

Embeddings convert text into vectors so that semantically similar text can be found using vector search.

Vector databases such as pgvector store embeddings and support similarity search.

Ollama can run local models like llama3.2 for chat and nomic-embed-text for embeddings.

In a RAG application, a PDF is read, split into chunks, converted to embeddings, stored in a vector database, retrieved based on similarity, and finally sent to the LLM as context.